

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 4

Formal Systems and the Architecture of Proof

Parallel Reading

Self-Study Curriculum

Parallel Reading: PCM, Part II (“The Origins of Modern Mathematics”), sections II.6 (“The Development of the Idea of Proof”) and II.7 (“The Crisis in the Foundations of Mathematics”), pp. 129-156, together they are the historical context for why these questions were asked at all.

Also read (essential paper): Kurt Godel’s incompleteness results. Do NOT read the original paper. Read: Nagel & Newman, *Godel’s Proof* (revised ed., 2001), chapters 1-7 (about 80 pages, very readable). If unavailable, read the PCM article on Godel’s theorem (section V.15, “Gödel’s Theorem”).